

PSimPy - Predictive and Probabilistic Simulation

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Want to build unified, robust and sustainable UQ workflows? ^[1]

PSimPy



Highlights

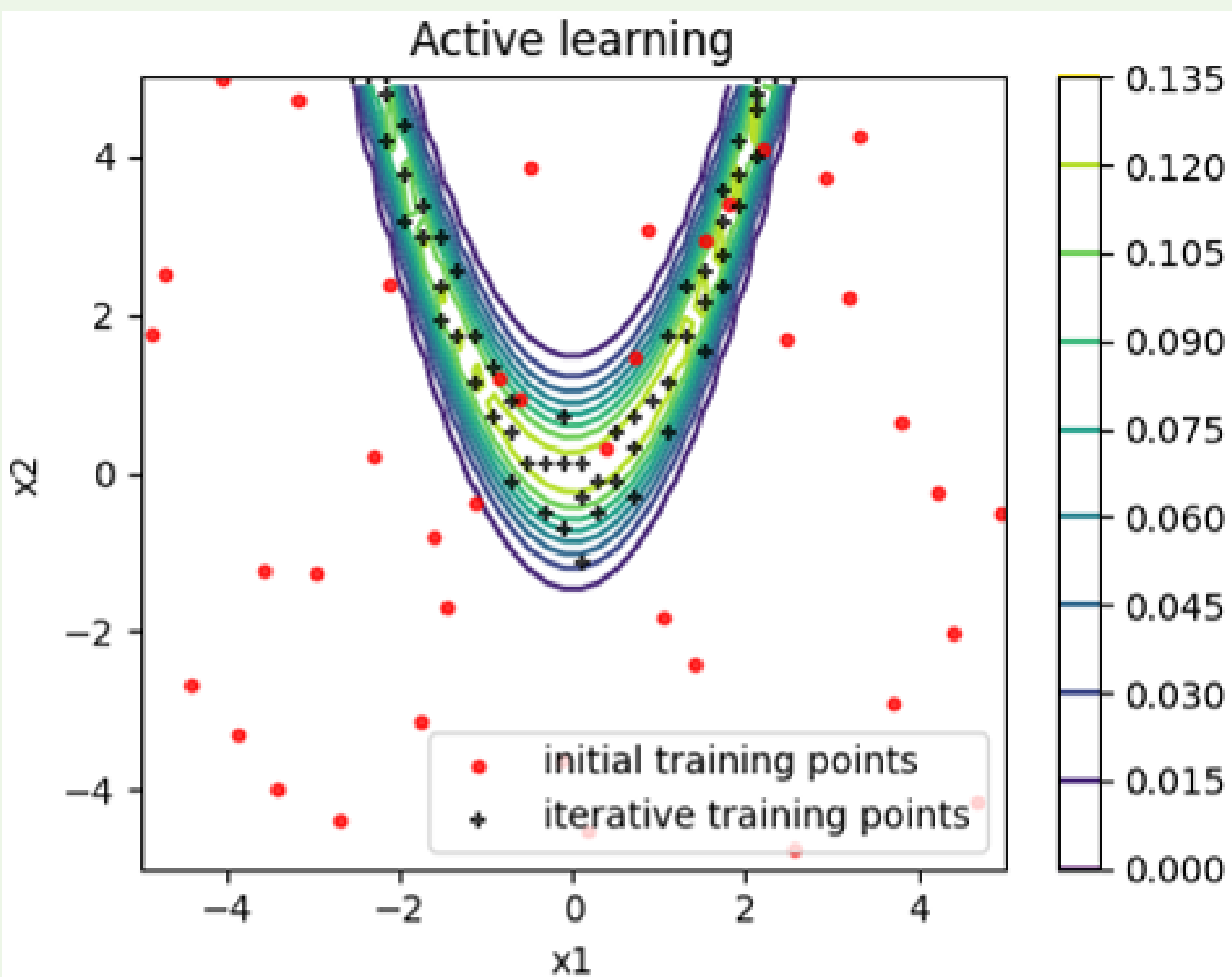
Robust GP emulator for your simulation.
Save precious compute time!

	Training	Evaluation	2500 Evals
Simulator	-	32 min	56 days
Emulator	4.5 days	0.05 sec	2 min

- Gaussian process emulation
- Bayesian calibration
- Global sensitivity analysis
- Efficient sampling
- Active learning

In Action

Quantify uncertainty, identify key drivers and calibrate with data.



Open source and Pythonic.

[1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, “PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators,” in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.